



Faster armies

Lightweight war chariots enabled armies to maneuver much faster than an infantry ever could. A Bronze Age chariot (c. 1200 BCE) moved more than 10 times as fast as the marching pace of a Roman legionary.

Egyptian chariot

Around 1600 BCE, the Egyptians developed lightweight war chariots that had spoked wheels and a thin wooden semicircular frame. The platform could accommodate two people, one to maneuver at high speed, and another armed with a bow.

wood bent into V-shape to make spokes

hub or nave

cattle intestines fasten spokes to hub

footboard made of sycamore wood

leather bindings connect shaft to chariot body

“...BUT LET THE LEFT-HAND HORSE KEEP SO CLOSE IN THAT THE NAVE [HUB] OF THE WHEEL SHALL ALMOST GRAZE THE POST.”

Homer, Greek poet, from *Iliad* Book XXIII, first description of chariot race, c.750 BCE

Neolithic period

Logroller

Neolithic people place loads on rollers made from logs. These logs, however, are not always smooth and the difficulties in keeping them aligned make this an inefficient method.



Early logrollers

c.1323 BCE

Spoked wheel

Wheels with spokes are lighter than disk wheels, and allow a cart or war chariot to be pulled by a lighter animal, such as a horse. First developed in the steppes of central Asia a little after 2000 BCE, these wheels spread to Egypt by 1600 BCE.

c.750 BCE

Iron-rimmed wheel

The Celts add iron rims to the wooden wheels of chariots to improve their durability on rough surfaces. They do so first by nailing the metal to the rim and later, by applying strips of hot iron, which shrink to fit as they cool.



Celtic chariot

3500 BCE

Potter's wheel

In southern Mesopotamia, potters become the first to use wheels to mechanize an industrial process—that of making pottery. They use a heavy, rapidly turning stone wheel to shape clay on.



Egyptian potter

c.2500 BCE

Disk wheel

The first true transportation wheels—disks of wood connected by axles—are developed in the Balkans and Mesopotamia. The Sumerians used these on battle wagons.

Disk wheels on the Standard of Ur



c.300 BCE

Water wheel

The Greeks invent water wheels as a means of harnessing the power of running water. They use water wheels either to raise water in buckets to a higher level for irrigation, or to drive around a shaft that operates a milling machine.